Astrobee



2024

Kibo-RPC 2024

“Astrobee, A free flying robot on ISS”

Jose Benavides, Lead, Autonomous Systems and Robotics

Jose Benavides Background

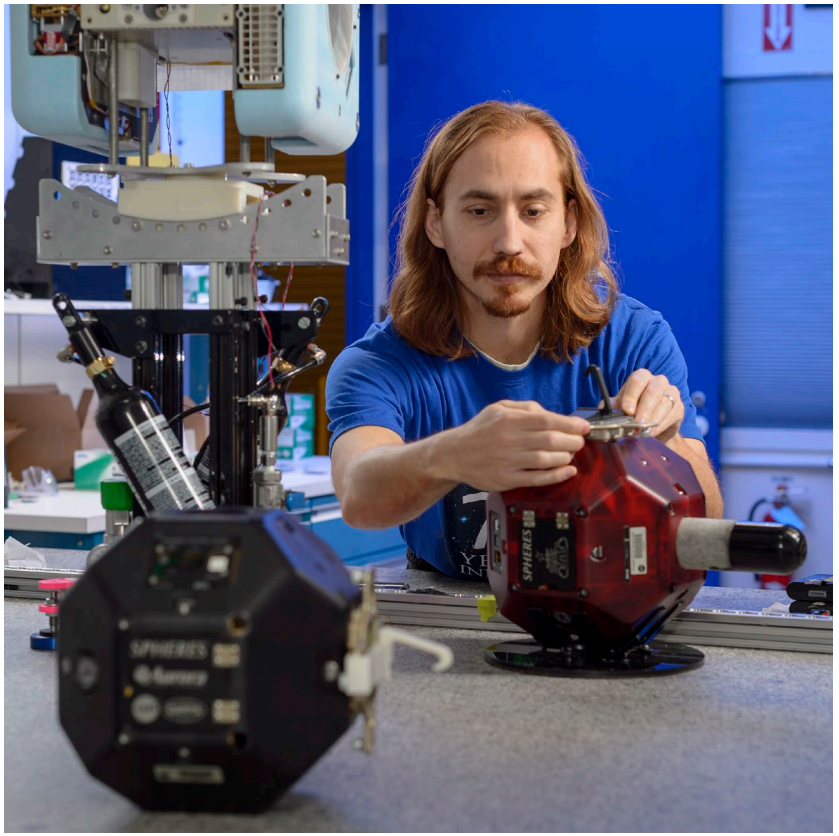
- Grew up in Chandler, AZ. My parents were raised as migrant farm workers and were the first to get college degree's
- Jose did several Internships with Microchip Technology and IBM
- He has a bachelors and masters degree from Arizona State University in Electrical Engineering, with specialization in control systems. His hobbies include reading, hiking, dancing, and computer networking.





Background Cont.

- Jose started employment as a contractor at NASA's Ames Research Center California's Silicon Valley in 2008 as a member of the Advanced Control and Evolvable Systems (ACES) Group in the Intelligent Systems Division.
- His past projects include SPHERES, rapid prototyping of controls systems using FPGAs, controls-oriented vehicle design of air-breathing hypersonic aircraft, trajectory generation and optimization for tactical flight management systems, pilot-in-the-loop high fidelity 6-dof flight simulation, and small satellite software. His research interests include embedded systems, rapid prototyping of control systems, spacecraft, small satellites, robotics, and human-machine interaction (HMI).
- Jose was recently the Project Manager of the NASA International Space Station (ISS) Astrobee Facility project. In conjunction with the prior platform, SPHERES. the [Astrobee Facility is one of the most used and popular ISS National Lab Facilities with over 80 on-board test sessions and 400+ hours of Facility Console activities involving crew.
- Jose is currently the Technical Area Lead for Autonomous Systems and Robotics at NASA Ames Research Center



Autonomous Systems and Robotics (ASR) Executive Summary

Capabilities:

- Enabling complex systems and operations through the development of technologies that can respond or adapt to changing conditions, knowledge, and constraints
- Creating automated and autonomous behaviors via robotic assistants, robust avionics, intelligent planning and scheduling, and advanced control technologies
- Agency leadership in applying autonomy and robotics expertise to NASA missions and terrestrial demonstrations

Products/Deliverables:

- Avionics software
- Adaptive and optimal control and estimation
- Advanced flight management systems
- Automated planning and scheduling
- Control agent architectures
- Computer vision & digital mapmaking
- Distributed and multi-agent systems
- Drilling automation
- Metamaterials and self-assembling structures
- Space robotics (planetary rovers & free-flyers)

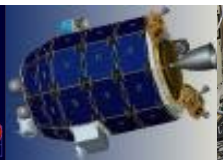
Customers/Projects/Partners: ASR Executes over 30 projects for various mission directorates and external partners.

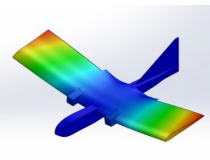
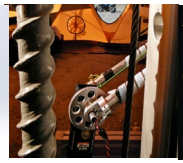
ARMD Projects -- Performance Adaptive Aeroelastic Wing; Autonomous Systems Distributed Sensing; small UAS Autonomy; Automation Enabled Pilot
SMD Projects -- Volatiles Investigation Polar Exploration Rover; Drilling and Sampling Automation
STMD Projects – Distributed Spacecraft Autonomy; Automated Reconfigurable Mission Adaptive Digital Assembly Systems (ARMADAS); Integrated System for Autonomous and Adaptive Caretaking (ISAAC), ; Tensegrity Robots;
ESMD/SOMD Projects --Autonomous Systems and Operations Vehicle System Management/Gateway; Human Landing Systems; Astrobee Facility ISS ops
External --Nissan Self-Driving Car, USGS, SFO

Resources: Greater than 32 Million Dollars
Workforce: ~40
Facilities/Labs:

1. Advanced Control Technology Lab
2. Coded Structures Lab
3. Astrobee Facility
4. NASA UAS Research Complex
5. Robotics labs and Roverscape



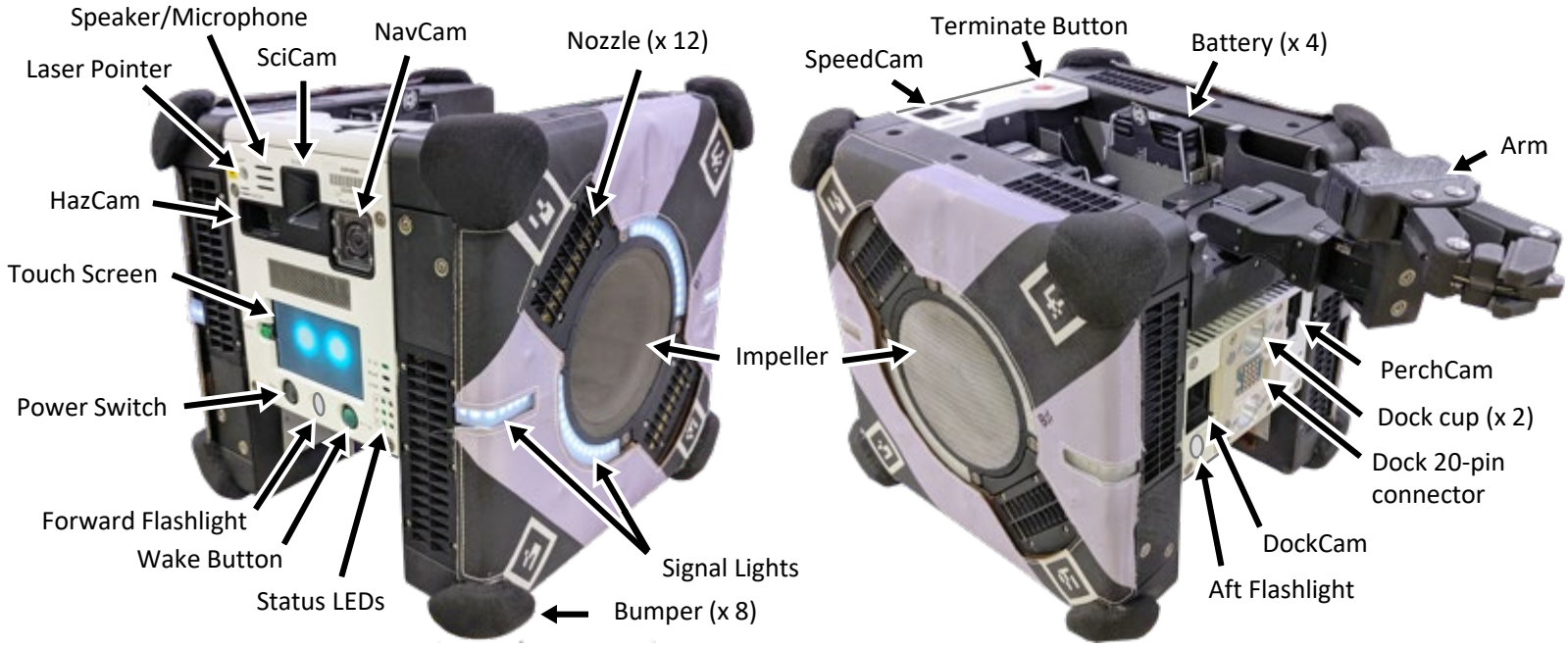




Astrobee Objectives

- Provide a microgravity robotic research facility in the ISS US Orbital Segment (USOS), which will replace the existing SPHERES facility
- Provide remotely operated mobile camera views of the ISS USOS to enhance the situation awareness of mission control
- Perform mobile sensor tasks in the ISS USOS



System Description - Hardware

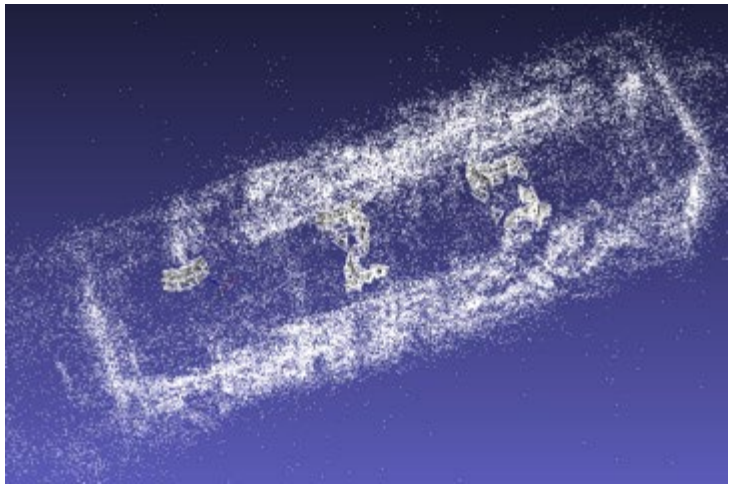


- Free flying robot inside the ISS
- 32 cm wide, ~9.1kg (2 batt., no arm)
- All electric + fan-based propulsion
- Robot arm for “perching”, ~1kg

- Three smartphone computers
- Three payload bays for expansion
- Microphone not currently enabled

System Description - Navigation

- Vision-based navigation
  - Compares features with on-board a priori map
  - Incorporates inertial measurements
- Fiducials used for autonomous docking
  - Requires approximately 1 cm position accuracy
- Visual odometry
  - Robot can continue to navigate where no map features are recognized

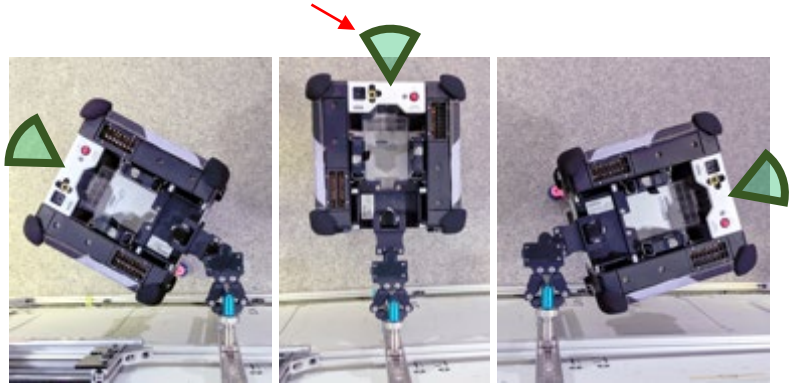


Feature map of the JEM-PM

System Description - Perching Arm

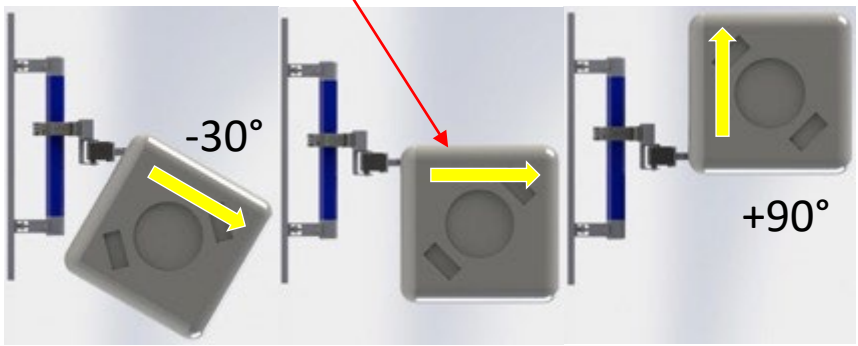
- Designed to grasp handrails
- Stows completely in payload bay
- Acts as a pan-tilt unit while perched
- Flexible and back-drivable
- May be perched manually

Camera View Direction



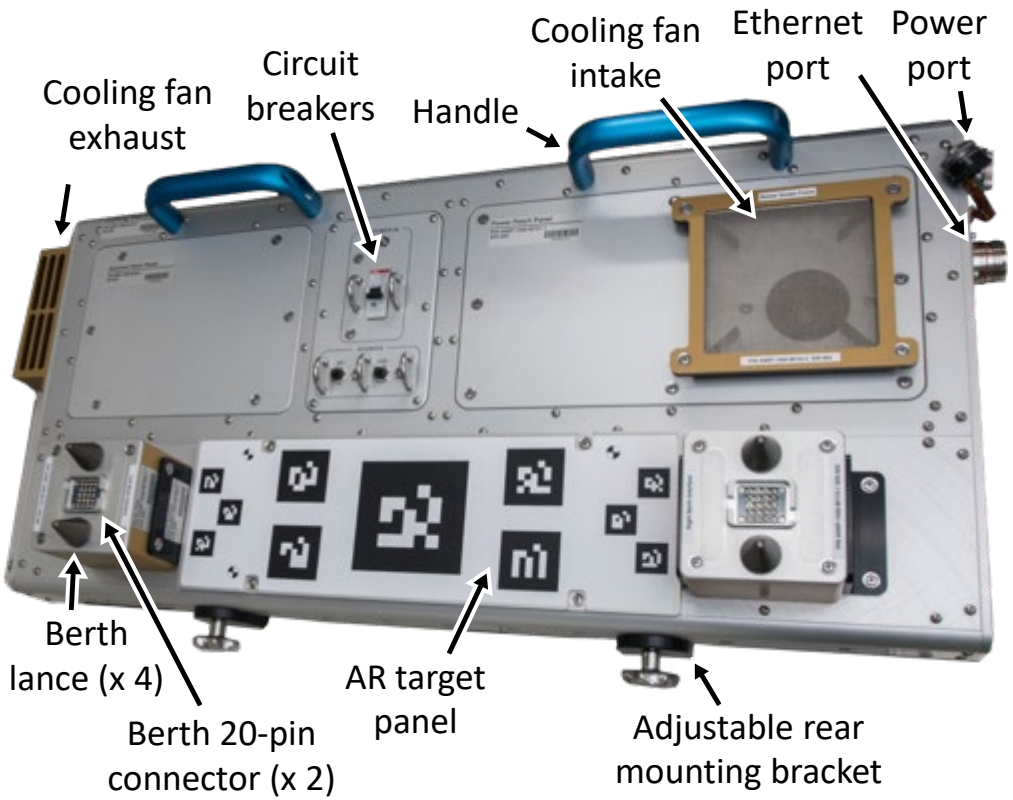
Astrobee Perching Arm pan motion

Camera View Direction



Astrobee Perching Arm tilt motion

System Description - Docking Station



- 85 cm x 38 cm x 28 cm
- Berths for 2 free flyers
- Provides power and Ethernet

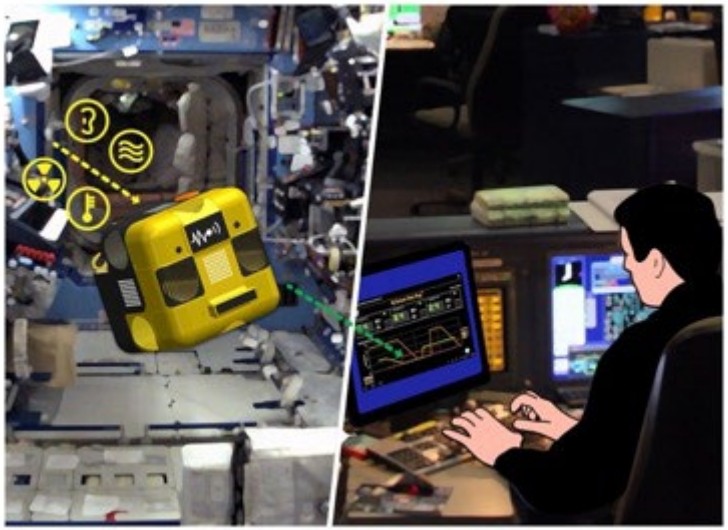
- Fiducials used for visual servoing to autonomously dock
- Magnets provide retention force

ISS Crew helpers

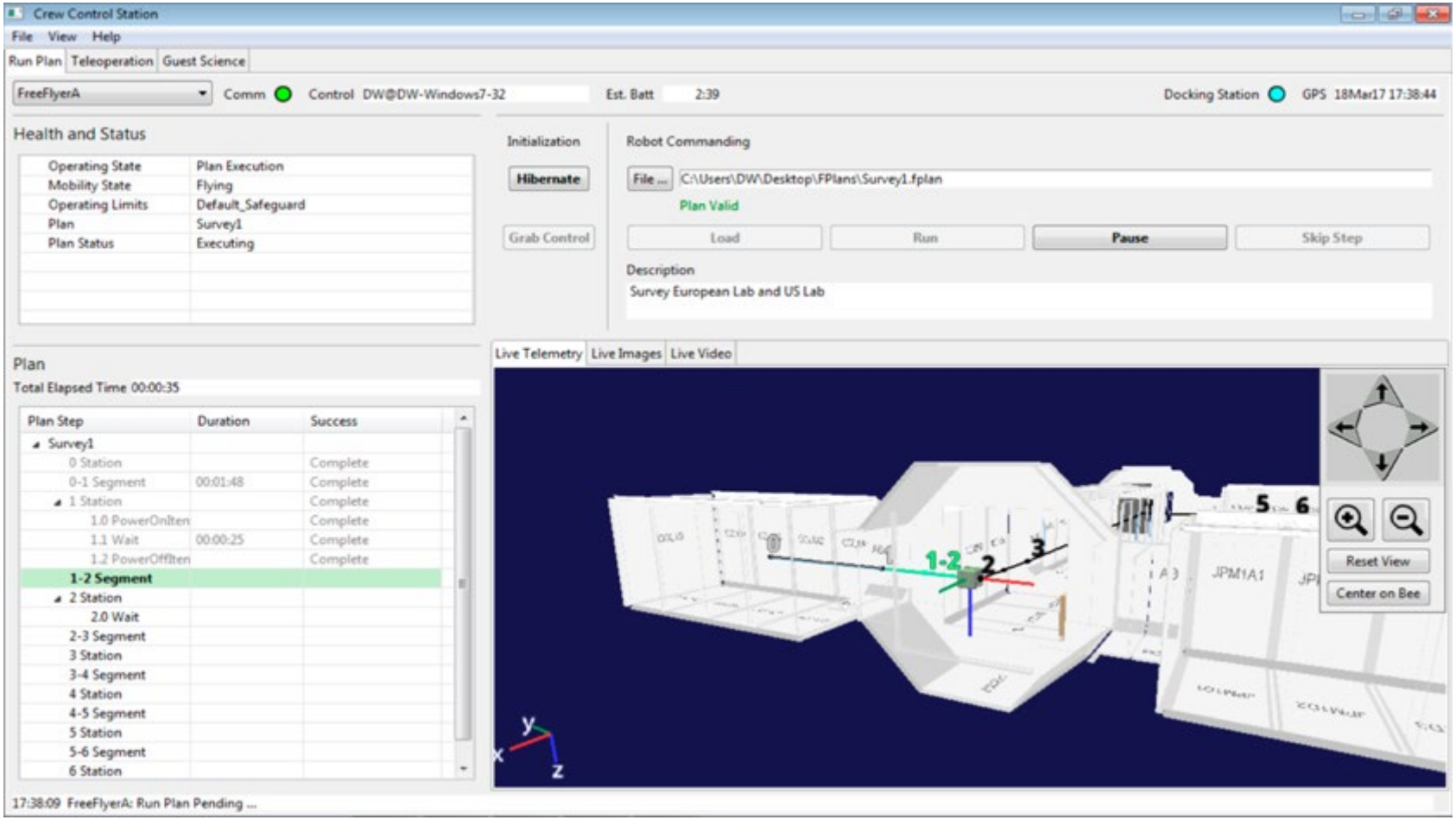


System Description - Ground Data System

- Astrobee Control Station
  - Sortie planning tool
  - Execution monitoring
    - Live telemetry
    - Image and video streams
    - 3D virtual display
  - Supervisory control (run plans or single commands)
  - Typically used by ground operators
- Crew Control Station runs on an EXPRESS Laptop Computer (ELC)
- Server for archiving and distributing Astrobee data
- Suite of engineering tools to support maintenance and software upgrades



Astrobee Control Station



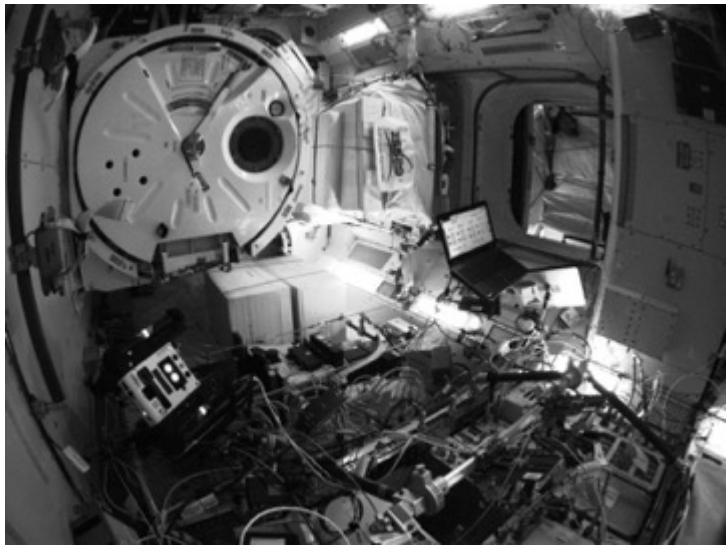


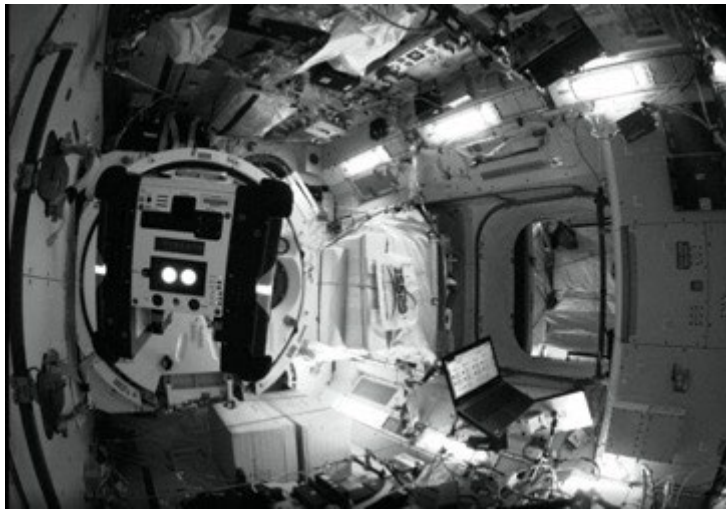
Basic Conops

- When an Astrobee is idle, it charges in its dock
- Astrobees can execute complex plans with full autonomy and no astronauts present
  - Including undock, traverse multiple modules, return to dock
- However, Astrobees run with ground operator oversight
  - When an anomaly occurs, an Astrobee generally stops and waits for operator intervention
  - It can continue operating during communication outages until it encounters an anomaly
- The operator can always take over and teleoperate
- Astronauts can also be operators, but this is will likely be a rare occurance (minimize crew time)

Perching Demonstration

- Honey perched autonomously on 07/26/2021
- Perch location was also useful to see how Bumble had become entangled.
- Operator was able to use the view from Honey to free Bumble.





Completed Investigations

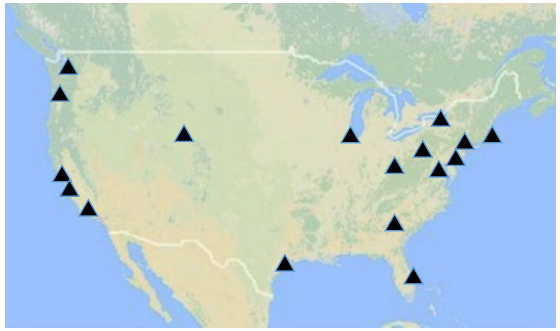
- Kibo Robotic Programming Competition – Year 1 – Completed 10/2020
- The REduced Gravity Gecko Adhesion docking Experiments (REGGAE) – Completed 01/2021
- Stanford Gecko – Phase 1 – Completed 04/2021



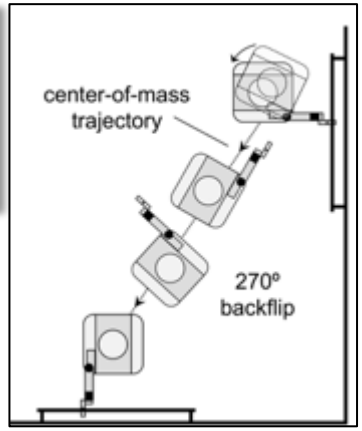


Ongoing Investigations

- Astrobatics (Naval Postgraduate School)
- SoundSee (Astrobotic/Bosch)
- Gecko (Stanford)
- RFID Recon (NASA AES/REALM-2)
- JAXA Kibo-RPC
- Astroporter (Tethers Unlimited)
- ISAAC (NASA STMD/GCD)
- ROAM (MIT/DLR)
- ReSWARM
- SVGS (FIT)
- SOARS (Zero-g Horizons)







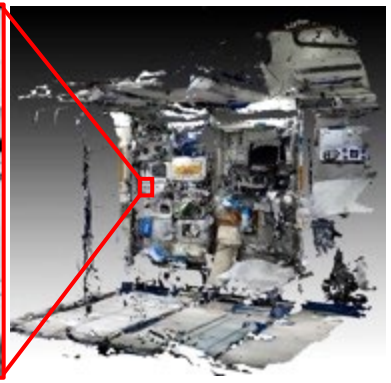
Achievements

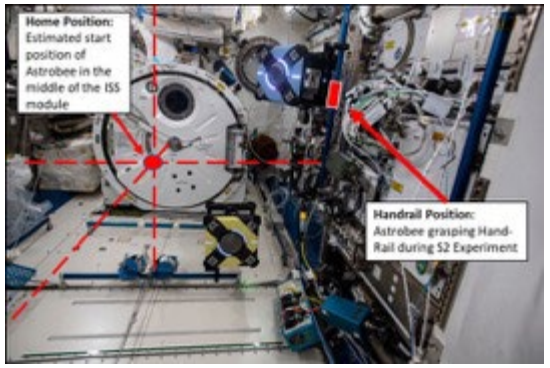


- 3 hours of continuous operation without crew intervention
- One Astrobee unit used to rescue a second
- 20 user sessions
- On-orbit repair of dock
- Autonomous Perching











(a)



(b)



(c)



(d)

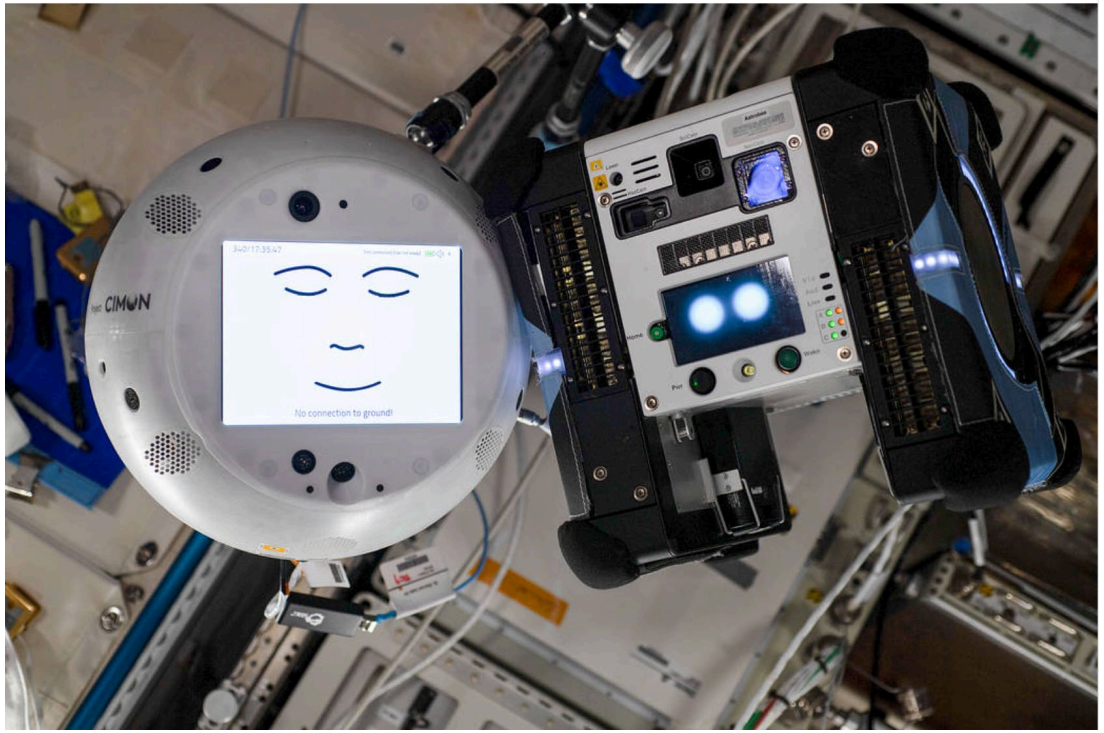
Robot Colleagues

JAXA's Int-Ball



Mar 22, 2022

Robot Colleagues Meet Aboard the International Space Station




QUESTIONS?